

SAVANNA AGCM, AB, Canada

WMO: 715470

Lat: 56.0761N

Lon: 119.3447W

Elev: 703

StdP: 93.16

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-30.9	-28.3	-34.4	0.2	-30.8	-31.6	0.2	-28.1	13.2	2.0	12.1	1.4	1.9	270	0.705

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	27.0	15.5	25.1	14.8	23.6	14.2	17.2	23.6	16.2	22.7	15.3	21.6	4.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.8	11.5	19.8	13.7	10.7	18.1	12.7	10.0	17.4	51.0	23.6	47.9	22.8	45.2	21.7	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.4	8.3	-34.1	29.7	2.7	1.8	-36.0	31.0	-37.6	32.0	-39.1	33.0	-41.0	34.3		
(5)			-34.3	18.9	2.6	1.9	-36.1	20.3	-37.7	21.4	-39.2	22.5	-41.1	23.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-10.3	-10.7	-5.9	2.8	10.3	14.0	16.1	15.2	10.3	3.2	-6.6	-11.8
	DBStd	12.08	9.26	8.47	7.85	5.00	4.40	3.11	2.67	3.36	4.56	5.04	7.37	8.06
	HDD10.0	3422	629	579	493	221	49	4	1	3	50	217	498	677
	HDD18.3	5878	887	812	752	467	248	133	77	107	242	470	748	936
	CDD10.0	606	0	0	0	4	59	124	191	163	59	4	0	0
	CDD18.3	22	0	0	0	0	1	3	9	8	1	0	0	0
	CDH23.3	403	0	0	0	2	30	51	153	136	32	0	0	0
	CDH26.7	62	0	0	0	0	3	3	24	28	3	0	0	0
	WSAvg	3.6	3.8	3.5	3.5	4.0	4.1	3.7	3.3	3.2	3.5	3.9	3.5	3.6
	PrecAvg	459	30	22	26	21	34	73	79	51	33	28	30	31
(15) Precipitation	PrecMax	655	67	45	69	46	86	153	146	113	83	94	88	129
	PrecMin	288	10	1	6	4	8	26	28	8	8	7	6	4
	PrecStd	98	15	13	14	10	20	38	29	28	18	19	18	25
	PrecStd	98	15	13	14	10	20	38	29	28	18	19	18	25
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	6.3	9.8	20.2	26.7	27.0	29.1	29.7	26.5	19.3	8.5	5.7
		MCWB	2.5	2.8	4.8	9.9	12.4	15.3	16.8	16.4	14.8	10.9	4.7	2.2
	2%	DB	5.3	4.6	7.5	15.4	23.2	24.8	26.8	26.8	23.8	15.1	5.9	4.1
		MCWB	2.3	1.8	3.2	7.5	11.4	14.1	16.0	15.9	13.7	8.7	3.0	1.2
	5%	DB	4.2	3.0	5.5	12.2	20.8	23.2	25.0	24.4	20.8	12.4	4.4	2.0
		MCWB	1.7	0.7	2.0	5.6	10.3	13.6	15.4	15.2	12.6	7.1	1.7	-0.4
	10%	DB	2.8	1.3	3.9	10.0	18.8	21.3	23.1	22.5	18.1	10.4	3.1	0.0
		MCWB	0.6	-0.7	0.9	4.5	9.5	12.7	14.7	14.4	11.3	5.7	0.7	-2.0
	0.4%	WB	3.2	3.2	5.2	10.6	14.0	17.1	19.5	18.3	16.1	11.6	5.4	2.8
		MCDB	5.6	5.7	8.9	19.8	22.2	24.4	26.6	24.6	24.1	16.9	7.6	5.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	2.5	1.9	3.3	8.0	12.3	15.7	17.4	17.1	14.4	9.3	3.2	1.3
		MCDB	4.9	4.3	6.9	14.2	20.6	22.1	23.4	23.7	21.8	14.0	5.4	4.0
	5%	WB	1.8	0.8	2.1	6.0	11.1	14.6	16.4	16.1	13.1	7.6	2.0	-0.3
		MCDB	4.2	2.9	5.1	11.4	19.2	21.0	22.6	22.8	20.0	11.6	4.0	1.8
	10%	WB	0.6	-0.7	1.0	4.7	10.2	13.5	15.4	15.1	11.8	6.1	0.9	-2.2
		MCDB	2.9	1.3	3.8	9.5	17.6	19.5	21.4	21.2	17.3	9.8	3.0	-0.2
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	8.2	9.1	9.6	12.0	10.9	11.4	11.8	11.2	8.1	7.6	7.8
		MCDBR	7.5	8.1	9.9	13.2	15.4	14.1	15.1	15.2	15.4	11.8	8.6	8.7
		MCWBR	5.8	6.0	6.7	7.2	6.9	6.0	6.7	6.6	7.1	6.8	6.7	7.3
		MCDBR	7.3	8.5	9.4	12.8	13.8	12.3	13.5	13.8	15.0	11.3	8.8	9.0
	5% WB	MCWBR	5.8	6.6	6.6	7.3	6.5	5.8	6.9	6.7	7.3	7.1	7.3	7.7
		MCWBR	5.8	6.6	6.6	7.3	6.5	5.8	6.9	6.7	7.3	7.1	7.3	7.7
(40) Clear-Sky Solar Irradiance	taub		0.250	0.250	0.292	0.317	0.341	0.344	0.345	0.343	0.304	0.281	0.238	0.220
	taud		2.216	2.336	2.281	2.395	2.402	2.407	2.432	2.427	2.544	2.559	2.410	2.239
	Ebn at Noon		669	844	878	907	899	898	890	869	867	807	713	630
	Edh at Noon		53	70	97	101	107	108	104	98	75	58	43	39
(44) All-Sky Solar Radiation	RadAvg		0.70	1.66	3.16	4.70	5.58	6.03	5.86	4.76	3.27	1.73	0.77	0.48
	RadStd		0.07	0.13	0.24	0.25	0.38	0.49	0.35	0.42	0.28	0.20	0.07	0.04

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page